

PE-EC801A — Antennas and Propagation

Semester VIII • 3L:0T:0P • 3 Credits

Module 6 Exam Notes

CGEC | B.Tech ECE (2023–27) | *Biraj Sarkar*

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1. Uniform Linear Arrays (ULA)

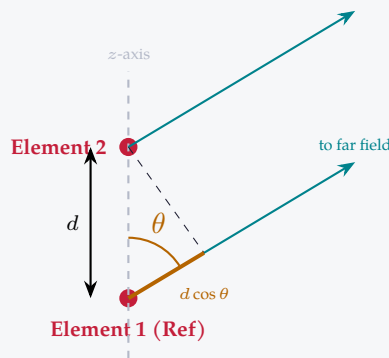
Definition of an Antenna Array

An **antenna array** is a set of two or more identical antenna elements positioned and excited such that their individual radiation fields combine constructively in desired directions and cancel destructively in other directions, increasing directivity.

1.1 Two-Element Array Analysis

Consider two identical antennas separated by a distance d along the z -axis. Antenna 1 is at the origin and serves as the phase reference ($I_1 = I_0 e^{j0}$). Antenna 2 is fed with a phase lead β ($I_2 = I_0 e^{j\beta}$).

Two-Element Array Geometry



In the far field, the path difference between the waves is $d \cos \theta$ (where θ is measured from the array axis). The total phase difference ψ between the fields is:

$$\psi = kd \cos \theta + \beta$$

The total far-field electric field is:

$$E_{total} = E_1 + E_2 = E_0 (1 + e^{j(kd \cos \theta + \beta)}) = E_0 e^{j\psi/2} \left(2 \cos \frac{\psi}{2} \right)$$

The normalized **Array Factor (AF)** is:

$$AF_n(\theta) = \cos \left(\frac{kd \cos \theta + \beta}{2} \right)$$

1.2 N-Element Uniform Linear Array (ULA)

For N identical elements fed with equal amplitude I_0 and progressive phase shift β :

$$AF = \sum_{n=0}^{N-1} e^{jn(kd \cos \theta + \beta)} = \sum_{n=0}^{N-1} e^{jn\psi}$$

Summing this geometric progression:

$$|AF| = \left| \frac{\sin \left(\frac{N}{2} \psi \right)}{\sin \left(\frac{1}{2} \psi \right)} \right| \Rightarrow |AF_n| = \left| \frac{\sin \left(\frac{N}{2} (kd \cos \theta + \beta) \right)}{N \sin \left(\frac{1}{2} (kd \cos \theta + \beta) \right)} \right|$$

1.3 Broadside vs. End-Fire Arrays

1. Broadside Array: Maximum radiation is perpendicular to the array axis ($\theta = 90^\circ$).

- *Condition:* $\psi = 0$ at $\theta = 90^\circ \Rightarrow kd \cos(90^\circ) + \beta = 0 \Rightarrow \beta = 0$.
- *Nulls:* Occur when $\sin(N\psi/2) = 0 \Rightarrow \theta_{null} = \cos^{-1}(\pm \frac{m\lambda}{Nd})$, $m = 1, 2, 3, \dots$

2. End-Fire Array: Maximum radiation is along the array axis ($\theta = 0^\circ$ or 180°).

- *Condition for $\theta = 0^\circ$:* $\psi = 0 \Rightarrow kd \cos(0^\circ) + \beta = 0 \Rightarrow \beta = -kd$.
- *Condition for $\theta = 180^\circ$:* $\psi = 0 \Rightarrow kd \cos(180^\circ) + \beta = 0 \Rightarrow \beta = kd$.

1.4 Key ULA Performance Formulas

For a uniform linear array of N elements with spacing d :

- **Broadside HPBW:**

$$\text{HPBW} \approx \frac{0.886 \lambda}{Nd} \text{ rad} = \frac{50.8^\circ \lambda}{Nd}$$

The beamwidth narrows as N or d increases — doubling the array length halves the HPBW.

- **Broadside Directivity:**

$$D \approx \frac{2Nd}{\lambda} \quad (\text{for } d \leq \lambda/2)$$

- **End-Fire HPBW:**

$$\text{HPBW} \approx 2\sqrt{\frac{2\lambda}{Nd}} \text{ rad}$$

- **First Sidelobe Level (Uniform Excitation):** -13.5 dB below the main beam — independent of N .

2. Non-Uniform Amplitude Arrays

Uniform arrays have high directivity but suffer from relatively high sidelobes (-13.5 dB for the first sidelobe). To reduce sidelobes, elements can be excited with non-uniform current amplitudes.

2.1 Binomial Array

The excitation amplitudes of the elements correspond to the coefficients of a binomial expansion:

$$a_n = \binom{N-1}{n}, \quad n = 0, 1, \dots, N-1$$

For example, for a 5-element array, the amplitudes scale as $1 : 4 : 6 : 4 : 1$.

- **Advantage:** Completely eliminates sidelobes if the element spacing $d \leq \lambda/2$.
- **Disadvantage:** Broader beamwidth (HPBW) and lower directivity than a uniform array of equivalent length.

2.2 Dolph-Tschebyscheff Array

This array uses Tschebyscheff (Chebyshev) polynomials $T_m(z)$ to design a pattern with a specified sidelobe level (SLL) and the narrowest possible main beamwidth. The Chebyshev polynomials are defined by:

$$T_m(z) = \cos(m \cos^{-1} z) \quad (|z| \leq 1)$$

$$T_m(z) = \cosh(m \cosh^{-1} z) \quad (|z| > 1)$$

This method optimizes the trade-off by ensuring that all sidelobes are at the exact same specified level (e.g., -20 dB or -30 dB).

Design Procedure:

1. Specify the number of elements N (array order $m = N - 1$) and the desired sidelobe level ratio R (e.g., $R = 26.3$ for -20 dB SLL; $R = 10^{SLL_{dB}/20}$).
2. Compute the parameter z_0 that maps the main lobe maximum to $T_m(z_0)$:

$$z_0 = \cosh \left[\frac{1}{m} \cosh^{-1}(R) \right]$$

3. Substitute $z = z_0 \cos(\frac{\psi}{2})$ into $T_m(z)$ to obtain the array polynomial. The AF takes the form $AF(\psi) = T_m(z_0 \cos \frac{\psi}{2})$.
4. Extract the excitation coefficients a_n by expanding the Chebyshev polynomial in powers of $\cos^n(\frac{\psi}{2})$.

Key result: For a 5-element array with -20 dB SLL, the excitation amplitudes are approximately $1 : 2.41 : 3.14 : 2.41 : 1$ (compare to binomial $1 : 4 : 6 : 4 : 1$ — Dolph gives a narrower beam for the same SLL).

3. Extension to Planar Arrays

A **planar array** consists of elements positioned in a two-dimensional grid (e.g., M elements along x with spacing d_x , and N elements along y with spacing d_y).

The overall Array Factor is the product of the individual linear array factors along the two directions:

$$AF_{planar} = AF_x \cdot AF_y = \left[\frac{\sin(\frac{M}{2}\psi_x)}{\sin(\frac{1}{2}\psi_x)} \right] \left[\frac{\sin(\frac{N}{2}\psi_y)}{\sin(\frac{1}{2}\psi_y)} \right]$$

where:

$$\psi_x = kd_x \sin \theta \cos \phi + \beta_x$$

$$\psi_y = kd_y \sin \theta \sin \phi + \beta_y$$

By dynamically adjusting the phase shifters β_x and β_y , the main beam can be steered to any coordinate (θ_0, ϕ_0) in 3D space:

$$\beta_x = -kd_x \sin \theta_0 \cos \phi_0$$

$$\beta_y = -kd_y \sin \theta_0 \sin \phi_0$$

4. Antenna Array Synthesis

Array synthesis involves finding the feeding currents (amplitudes and phases) and element locations required to produce a desired radiation pattern.

4.1 Schelkunoff Polynomial Method

Schelkunoff represented the array factor of an N -element linear array as a polynomial in the complex variable $z = e^{j\psi}$:

$$AF = \sum_{n=0}^{N-1} a_n z^n = a_{N-1}(z - z_1)(z - z_2) \dots (z - z_{N-1})$$

where z_i are the roots (zeros) of the polynomial. Since z lies on the unit circle ($|z| = 1$), the nulls in the radiation pattern correspond to the roots that lie on the unit circle.

Visible Region: As θ varies from 0° to 180° , $\psi = kd \cos \theta + \beta$ traces an arc on the unit circle in the complex z -plane. This arc is called the **visible region**:

$$-kd + \beta \leq \psi \leq kd + \beta$$

- For $d = \lambda/2$ and $\beta = 0$: visible region spans $-\pi \leq \psi \leq \pi$ (full circle).
- For $d < \lambda/2$: visible region is less than a full circle — some roots in the **invisible region** do not contribute to the radiation pattern.
- Placing a root *inside* the visible region creates a null in the pattern. Placing a root *outside* the visible region eliminates that null.

This allows designers to synthesize specific patterns by precisely controlling which roots lie on vs. off the visible arc.

4.2 Woodward-Lawson Method

The Woodward-Lawson method synthesizes a desired source pattern by superimposing several independent, orthogonal beams (each having a $\frac{\sin(N\psi/2)}{N \sin(\psi/2)}$ shape centred at different angles). The excitation coefficients are found by sampling the desired pattern at the beam centres.

Sampling formula: For N elements, the array factor is:

$$AF(\psi) = \sum_{m=-(N-1)/2}^{(N-1)/2} c_m \cdot \frac{\sin\left[\frac{N}{2}(\psi - \psi_m)\right]}{N \sin\left[\frac{1}{2}(\psi - \psi_m)\right]}$$

where each beam is centred at $\psi_m = \frac{2\pi m}{N}$ and the sampling coefficients c_m are found by evaluating the desired pattern $F_d(\psi)$ at those points:

$$c_m = F_d(\psi_m) = F_d\left(\frac{2\pi m}{N}\right)$$

Key properties:

- The synthesized pattern passes exactly through the desired sample values.

- Between sample points, the approximation may deviate (Gibbs-like ripple for abrupt transitions).
- Adding more elements N increases the number of sample points and improves accuracy.
- Suitable for pencil-beam, sector, flat-top, or shaped beam synthesis.

Quick Revision — Module 6

Antenna Arrays and Synthesis | PE-EC801A

Key Formulas and Metrics at a Glance

N-Element ULA Array Factor (AF)	$AF = \frac{\sin(N\psi/2)}{\sin(\psi/2)}$ where $\psi = kd \cos \theta + \beta$
Broadside Phase Shift (β)	$\beta = 0$
End-Fire Phase Shift (β)	$\beta = \mp kd$ (for $\theta = 0^\circ$ or 180° main beam)
Broadside HPBW	$HPBW \approx \frac{0.886\lambda}{Nd} \text{ rad} \approx \frac{50.8^\circ \lambda}{Nd}$
Broadside Directivity	$D \approx \frac{2Nd}{\lambda}$ (uniform excitation, $d \leq \lambda/2$)
Binomial Array Coefficients	$a_n = \binom{N-1}{n} = \frac{(N-1)!}{n!(N-1-n)!}$
Planar Beam Steering Phase	$\beta_x = -kd_x \sin \theta_0 \cos \phi_0, \quad \beta_y = -kd_y \sin \theta_0 \sin \phi_0$
Schelkunoff Complex Variable (z)	$z = e^{j\psi} = e^{j(kd \cos \theta + \beta)}$

One-Line Definitions

- **Array Factor:** The radiation pattern of an array of isotropic elements, describing how the geometry and feeds affect the pattern.
- **Grating Lobe:** An unwanted duplicate main beam of radiation that occurs when element spacing $d \geq \lambda$.
- **Binomial Array:** An array with non-uniform excitation coefficients that completely eliminates sidelobes.
- **Phased Array:** An array where the phase of each element is dynamically adjusted to steer the radiation beam without moving the antenna.
- **Woodward-Lawson Method:** A pattern synthesis technique based on the superposition of orthogonal beam patterns.

Frequently Asked / Viva Questions

Q1. Define Array Factor (AF). State the Principle of Pattern Multiplication.

The Array Factor is the radiation pattern of the array assuming isotropic radiating elements. The Principle of Pattern Multiplication states that the total far-field pattern (E_{total}) of an array of identical non-isotropic elements is the product of the single-element field pattern (E_{single}) and the Array Factor (AF):

$$E_{total}(\theta, \phi) = E_{single}(\theta, \phi) \times AF(\theta, \phi)$$

Q2. Differentiate between broadside and end-fire arrays.

Frequently Asked / Viva Questions (contd.)

A broadside array radiates its maximum power perpendicular to the axis of the array ($\theta = 90^\circ$), which requires in-phase feeds ($\beta = 0$). An end-fire array radiates along the axis ($\theta = 0^\circ$ or 180°), which requires a progressive phase shift $\beta = \mp kd$.

Q3. What are grating lobes? How can they be avoided in array design?

Grating lobes are secondary maximum beams that have the same amplitude as the main beam. They represent a waste of energy in undesired directions. They can be avoided by keeping the element spacing smaller than a wavelength ($d < \lambda$, or $d < \lambda/2$ for steering).

Q4. Why are binomial arrays designed? What is their main drawback?

Binomial arrays are designed to completely eliminate sidelobes in their radiation pattern. Their main drawback is that they produce a wider main beam width (reduced resolution) and have lower directivity compared to a uniform array of the same length.

Q5. Explain the key concept of the Dolph-Tschebyscheff array design.

It uses Chebyshev polynomials to find the optimal balance between beamwidth and sidelobe level. It yields the narrowest possible main beam width for a specified, constant sidelobe level.

Q6. Describe the basic idea of the Schelkunoff polynomial method.

The array factor is written as a complex polynomial $AF(z) = \sum a_n z^n$. The nulls of the radiation pattern correspond to the roots of the polynomial that lie on the unit circle in the complex plane ($|z| = 1$). Moving these roots allows shaping the pattern.

Q7. How does a 2D planar array steer a beam in three dimensions?

By using two independent progressive phase shifts β_x and β_y along the x - and y -axes, the phase can be matched to create a flat constructive wavefront in any direction (θ_0, ϕ_0) .

Comparison of Linear Array Types

Parameter	Uniform ULA	Binomial Array	Dolph-Tschebyscheff
Excitation Amplitudes	Equal (1 : 1 : 1 : 1)	Binomial coefficients (1 : 3 : 3 : 1)	Calculated using Chebyshev polynomials
Sidelobe Level	High (first SLL ≈ -13.5 dB)	No sidelobes (SLL = $-\infty$)	Constant SLL (set by designer)
Directivity	High	Low (broad beamwidth)	Optimum (narrowest beam for SLL)